

Measuring Effective Diversification During Financial Crises

A Copula-Based Approach Using Tail Dependence and Spectral Entropy

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Abstract

Traditional diversification metrics based on correlation matrices often underestimate systemic risk during financial crises. This project proposes a tail dependence-based framework that combines GARCH filtering, copula modeling, and spectral analysis to estimate the effective number of assets that remain truly diversified under extreme market conditions.

Keywords

Portfolio diversification; Tail dependence; Clayton copula; GARCH filtering; Spectral entropy; Shannon entropy; Information theory; Dependence modeling; Financial crises; Effective number of assets.

1 Introduction

Diversification is one of the most widely used strategies for portfolio risk management. By combining assets with different risk-return characteristics, investors seek to reduce overall portfolio risk while maintaining expected returns. However, nominal diversification does not necessarily imply effective diversification. Assets that appear distinct under normal market conditions may exhibit similar behavior during periods of financial stress, leading to a substantial increase in portfolio risk.

A contemporary example can be observed in portfolios heavily concentrated in technology and artificial intelligence-related companies. Although such portfolios may contain a large number of individual assets, their performance is often driven by common risk factors, reducing the true level of diversification.

This challenge has attracted considerable attention in portfolio theory, leading to the development of several diversification measures. Among the most used is the Diversification Ratio proposed by Choueifaty and Coignard, which compares the weighted average volatility of individual assets with the overall portfolio volatility. Another notable measure is the Effective Number of Bets (ENB), which quantifies the number of truly independent sources of risk contributing to portfolio performance.

While these metrics provide valuable insights into portfolio concentration and diversification, they are typically based on covariance or correlation structures estimated under normal market conditions, often underestimating systemic risk and failing to capture the increase in dependence that often emerges during periods of market stress.

This limitation motivates the development of alternative approaches capable of capturing dependence structures during periods of financial stress.

This report proposes a copula-based framework to measure portfolio dependence under extreme market conditions. By focusing on tail dependence rather than linear correlation in order to provide a more realistic assessment of diversification and the effective number of assets that remain truly independent during market crises.

2 Data and Methodology

The methodology consists of five main stages: data processing, volatility filtering, dependence modeling, construction of tail dependence matrix and entropy-based diversification measurement. Each stage of the methodology is accompanied by a brief mathematical overview to provide the necessary context for understanding the construction of the proposed metric.

2.1 Data description

The analysis is based on daily adjusted closing prices obtained for a set of representative financial assets: State Street SPDR S&P 500 ETF Trust (SPY), Invesco QQQ Trust (QQQ), iShares 20+ Year Treasury Bond ETF (TLT), iShares iBoxx \$ Inv Grade Corporate Bond ETF (LQD), Vanguard Real Estate Index Fund ETF (VNQ) and Schwab US REIT ETF (SCHH).

These assets were selected to provide exposure to three major segments of the financial market. SPY and QQQ represent equity index funds, TLT and LQD represent fixed-income securities, and VNQ and SCHH represent real estate investment trusts (REITs).

Missing observations were removed and prices were aligned to ensure a consistent time series across all assets.

Logarithmic returns were computed according to:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right),$$

where P_t denotes the asset price at time (t). Log returns are commonly used in financial applications due to their additive properties and statistical convenience.

2.2 Volatility filtering

Financial return series typically exhibit volatility clustering, whereby periods of high volatility tend to be followed by further periods of high volatility, and periods of low volatility tend to persist over time. As a consequence, the variance of returns is not constant, violating the homoscedasticity assumption required by many statistical models.

Since copula estimation captures the dependence structure between assets rather than volatility dynamics, it is desirable to remove conditional heteroskedasticity from each return series before modeling dependence. This problem motivates the following definition:

Definition: (GARCH Model)

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is a time series model designed to capture the time-varying volatility commonly observed in financial returns. It assumes that the conditional variance evolves over time as a function of past shocks and past volatility.

Consider the return at time t , denoted by r_t , such that:

$$r_t = \mu + \epsilon_t, \quad (1)$$

where:

$$\epsilon_t = \sigma_t z_t \quad z_t \sim iid \quad t_\nu$$

the conditional volatility follows the GARCH(1,1) process:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (2)$$

where ω represents the long-run variance level, α measures the impact of recent shocks on current volatility, and β captures volatility persistence over time.

By estimating and removing conditional volatility effects, standardized residuals can be obtained and subsequently used for dependence modeling through copulas:

$$\hat{z}_t = \frac{\hat{\epsilon}_t}{\hat{\sigma}_t} \quad (3)$$

2.3 Copula framework

A copula is a multivariate function that links marginal distributions to a joint distribution. Its main advantage lies in separating the marginal behavior of individual variables from their dependence structure.

According to Sklar's theorem, any multivariate distribution can be expressed in terms of its marginal distributions and an associated copula. For two random variables X and Y with marginal distribution functions F_X and F_Y , the joint distribution can be written as:

$$F_{X,Y}(x, y) = C_{X,Y}(F_X(x), F_Y(y)), \quad (4)$$

Here, $C : [0, 1]^2 \rightarrow [0, 1]$ denotes a copula function. A valid copula satisfies the boundary conditions:

$$\begin{aligned} C(u, 0) &= C(0, v) = 0, \\ C(u, 1) &= u, \quad C(1, v) = v, \end{aligned}$$

and for all $u, v \in [0, 1]$, and is 2-increasing, ensuring that it defines a proper joint distribution function.

A copula describes the way variables move together after removing the effects of their individual distributions.

2.4 Uniform Margins

Since copulas model the dependence structure separately from the marginal distributions, the standardized residuals must be transformed into uniformly distributed variables on the interval $[0,1]$.

To achieve this, the empirical probability integral transform was applied to each series of standardized residuals. Let z_t denote a standardized residual and let n be the sample size. The transformed observations are defined as:

$$u_t = \frac{\text{rank}(z_t)}{n + 1}, \quad (5)$$

This transformation maps the standardized residuals onto the unit interval while preserving their dependence structure because of its monotonic behavior. The resulting uniform variables constitute the input required for copula estimation.

2.5 Clayton Copula

Among the different copula families available, the Clayton copula was selected due to its ability to capture lower-tail dependence. This feature is particularly relevant in financial applications, where diversification tends to deteriorate during periods of market stress and extreme losses may occur simultaneously across assets.

The bivariate Clayton copula is defined as:

$$C_\theta(u, v) = \left(u^{-\theta} + v^{-\theta} - 1\right)^{-1/\theta}, \quad \theta > 0, \quad (6)$$

where u and v are uniform random variables on the interval $[0, 1]$, and θ is the dependence parameter. Larger values of θ correspond to stronger lower-tail dependence between the variables.

2.6 Dependence matrix

A bivariate Clayton copula was fitted to each pair of assets using the transformed uniform observations. Let $\hat{\theta}_{ij}$ denote the estimated copula parameter for assets i and j . The corresponding lower-tail dependence coefficient is computed as:

$$\lambda_{ij} = 2^{-1/\hat{\theta}_{ij}}. \quad (7)$$

The collection of pairwise tail dependence coefficients is assembled into a symmetric matrix

$$\Lambda = \begin{cases} 2^{-1/\hat{\theta}_{ij}} & i \neq j \\ 1 & i=j \end{cases} \quad (8)$$

Each element measures the strength of joint extreme losses between two assets. The matrix provides a representation of portfolio dependence during periods of market stress, where traditional correlation measures may underestimate systemic risk.

2.7 Spectral Entropy and Effective Diversification

Once the tail dependence matrix has been constructed, the next step is to quantify how concentrated the dependence structure is across the portfolio. While the pairwise coefficients provide information about individual asset relationships, they do not directly indicate the overall level of diversification that remains during periods of market stress.

To address this issue, a spectral analysis of the tail dependence matrix is performed.

Let the eigen-decomposition of Λ be:

$$\Lambda = QDQ^\top,$$

where D is a diagonal matrix containing the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.

A spectrum dominated by a small number of eigenvalues indicates that tail dependence is concentrated in a few common risk factors, and a more uniform spectrum suggests a diversified dependence structure.

To quantify this concentration, the eigenvalues are normalized as:

$$p_i = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j},$$

such that:

$$\sum_{i=1}^n p_i = 1.$$

The resulting values can be interpreted as a probability distribution over the dependence spectrum.

The spectral entropy of the portfolio is then defined using Shannon's entropy:

$$H = - \sum_{i=1}^n p_i \ln(p_i).$$

Higher entropy values indicate that dependence is distributed more evenly across the spectrum, while lower values imply that dependence is concentrated in a small number of dominant factors.

Finally, the effective number of assets under crises is defined as:

$$\mathbf{ENAUC} = e^H, \tag{9}$$

and

$$\mathbf{ENAUC}\% = \frac{e^H}{\# \text{nominal assets}} \tag{10}$$

This quantity may be interpreted as the number of effectively independent assets that remain in the portfolio under extreme market conditions. A value close to the nominal number of assets suggests that diversification largely survives during crises, whereas a value close to one indicates that most assets behave as a single highly dependent component.

3 Results

3.1 Tail Dependence Matrix Analysis

To empirically evaluate the behavior of the portfolio under extreme market conditions, the pairwise lower-tail dependence coefficients (λ_{ij}) derived from the fitted Clayton copulas were computed according to Equation (7). The resulting tail dependence matrix Λ was compared against the traditional Pearson linear correlation matrix, which represents the average global co-movement between assets.

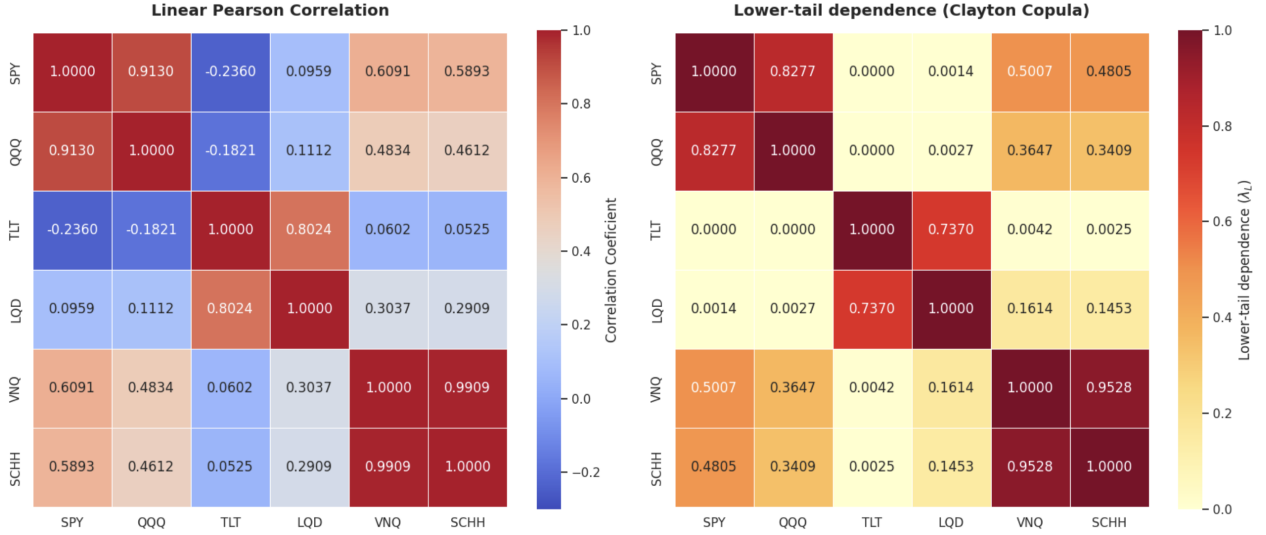


Figure 1: Comparison between Pearson Linear Correlation (Left) and Clayton Copula Lower-Tail Dependence (Right).

The empirical contrast between the two matrices reveals critical insights for systemic risk management and challenges traditional diversification assumptions:

- **SPY–QQQ remains the main risk factor:** The strong relationship between SPY and QQQ persists even under extreme market conditions, suggesting that a substantial portion of their co-movement is driven by common systemic risk factors rather than average market fluctuations alone.
- **SPY-TLT and QQQ-TLT:** The negative correlation already suggested a diversifying effect, but the lower-tail dependence shows something stronger: $\lambda_L = 0$, this suggest there is practically no contagion during extreme events. Treasury bonds appear largely disconnected from equity market crashes, preserving diversification benefits precisely when they are most needed.
- **TLT and LQD:** Despite representing different segments of the fixed-income market, TLT and LQD exhibit substantial lower-tail dependence, indicating that diversification within bond allocations may be weaker than expected during periods of stress.
- **VNQ and SCHH:** The real-estate ETFs VNQ and SCHH behave almost as a single risk factor, providing very limited diversification during market stress.

These findings demonstrate that average linear correlation and crisis tail dependence are not equivalent. While close relationships within the same sector remain strong under both measures (such as SPY-QQQ, VNQ-SCHH, and TLT-LQD), cross-asset dynamics change completely during extreme market down movements. Most notably, long-term Treasury bonds show a complete collapse in lower-tail dependence with equities ($\lambda_L = 0$), meaning that their negative average correlation does not translate into a joint drop during a crisis. Similarly, corporate bonds (LQD) decouple from equities, moving from a mild positive correlation to near-absolute tail independence. Ultimately, these results suggest that traditional correlation matrices can distort the true picture of systemic risk, either overstating or understating diversification benefits during a market panic.

3.2 Spectral Analysis

Table 1: Normalized Eigenvalue Spectrum		
Rank	Pearson Correlation	Tail Dependence
1	0.518	0.460
2	0.313	0.290
3	0.132	0.175
4	0.024	0.041
5	0.011	0.026
6	0.001	0.008

Table 1 presents the normalized eigenvalue spectra obtained from the Pearson correlation matrix and the proposed tail dependence matrix.

The largest eigenvalue accounts for 51.8% of the total dependence mass under Pearson correlation, indicating that a substantial fraction of portfolio risk is concentrated in a dominant common factor. Under the tail dependence framework, the first eigenvalue decreases to 46.0%, suggesting a less concentrated dependence structure during extreme market events.

Additionally, the third eigenvalue increases from 13.2% to 17.5%, indicating that a larger proportion of dependence is distributed across secondary factors. This result is consistent with the existence of multiple sources of crisis risk associated with different asset classes, including equities, fixed income securities, and real estate investment trusts.

Overall, the tail dependence spectrum appears more evenly distributed than the Pearson spectrum. Consequently, the corresponding spectral entropy is higher, resulting in a larger effective number of assets under crisis conditions.

3.3 Effective number of assets under crises

Table 2: Effective Diversification Metrics

Metric	Value
Nominal Number of Assets	6.00
ENA (Pearson Correlation)	3.07
ENAU (Tail Dependence)	3.62
Diversification Survival Ratio	60.3%

Although the portfolio contains six assets, the correlation structure suggests that only 3.07 independent sources of risk are effectively present. Under the proposed tail dependence framework, the Effective Number of Assets Under Crises (ENAU) increases to 3.62.

This result indicates that approximately 60% of the portfolio’s nominal diversification survives during extreme downside market conditions. In particular, Treasury bonds exhibit no lower-tail dependence with equity ETFs, allowing part of the diversification benefits to remain intact even during periods of market stress.

4 Conclusions

The proposed ENAUC framework aims to evaluate portfolio diversification under adverse market conditions by focusing on lower-tail dependence rather than average correlation. By combining GARCH filtering, copula-based dependence modeling and spectral entropy, the methodology seeks to quantify how much diversification effectively survives during periods of market stress.

The main limitation of the approach is that the tail dependence matrix is constructed from pairwise dependence estimates and is therefore not guaranteed to be positive semidefinite. In such cases, negative eigenvalues may arise, reducing the probabilistic interpretation of the spectral entropy. Although spectral clipping provides a practical solution, significant negative eigenvalue mass could lead to information loss and affect the stability of the metric.

Additional limitations include the use of a bivariate dependence framework and the exclusive focus on lower-tail dependence through the Clayton copula. Consequently, the metric is specifically designed for downside risk analysis and should not be interpreted as a complete description of the portfolio dependence structure.

Despite these limitations, the results suggest that ENAUC provides an intuitive and interpretable measure of crisis diversification. Future research could explore fully multivariate copula constructions, positive semidefinite dependence estimators, and analyze the evolution of diversification across different market regimes.

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